

JavaScript: *this*, the Event Argument, and `event.target.value`

My Question

I'm in a boot camp, full stack boot camp, front end and back end. I'm currently in the front end. I finished HTML, CSS, and I'm studying JavaScript. I'm in the middle of the way. So I want you to act as an instructor for me. I want you to give me information about sending `this` as an argument and also about event argument explanation very briefly and `event.target.value`.

The Answer

Great progress! Here's a short lesson on all three.

1. Passing *this* as an argument

In an inline HTML event, `this` refers to **the element itself**. Passing it to a function gives that function direct access to the element.

```
<button onclick="changeColor(this)">Click me</button>

<script>
  function changeColor(element) {
    element.style.backgroundColor = "orange";
    element.textContent = "Clicked!";
  }
</script>
```

Here, `element` is the button, so you can change its style, text, and so on.

2. The event argument

Whenever an event happens (click, keypress, input...), JavaScript creates an **event object** with details about what happened. With `addEventListener`, it's passed to your function automatically as the first parameter. People usually name it `event` or `e`.

```
const btn = document.querySelector("button");

btn.addEventListener("click", function (event) {
  console.log(event.type); // "click"
  console.log(event.target); // the element that was clicked
});
```

3. `event.target.value`

This is a chain of three parts:

- `event` is the event object
- `.target` is the element that triggered the event

- `.value` is the current value of that element (used with inputs, textareas, and selects)

```
<input type="text" id="name" placeholder="Type your name">
<p id="output"></p>

<script>
  const input = document.getElementById("name");
  const output = document.getElementById("output");

  input.addEventListener("input", function (e) {
    output.textContent = "Hello, " + e.target.value;
  });
</script>
```

Each time you type, `e.target.value` holds whatever is currently in the input box.

Quick tip: With a regular function, `this` inside the listener also equals the element. With an arrow function `() => {}`, it doesn't. That's one reason `event.target` is the safer habit.

Mini exercise

Create a `<select>` dropdown with three colors. When the user picks one, change the page's background to that color using `event.target.value`.